Academic Collaboration Network: System Simulation and Validation

Workshop No. 4 - Systems Analysis & Design 2026-I

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Abstract- This paper presents the computational simulation and validation phase of the Academic Collaboration Network, a socio-technical platform designed to reduce isolated learning through skill-based group optimization and institutional resource integration. Building on Workshops #1, #2 and #3, the study implements a process-oriented discrete-event model and a behavior-oriented agent-based network model. Results show that the optimized ACN scenario reduces median group formation time by 91.5%, reduces process-model isolated students by 67.4%, and maintains p95 matching latency at 1.24 seconds under a 500-student stress scenario. The B1 loop reduces final isolated students by 82.8% and increases density by 883.0%.

Keywords- systems engineering, discrete-event simulation, agent-based modeling, collaborative learning, social network analysis, design validation.

I. INTRODUCTION

The previous workshops established the Academic Collaboration Network (ACN) as a response to isolated learning, inefficient study group formation, and uneven distribution of academic support among university students. Workshop #1 identified the current coordination process as informal and fragmented: 88% of surveyed students relied exclusively on WhatsApp, 36% reported lack of communication as the main barrier, and 96% expressed positive or conditional interest in a dedicated platform.

Workshop #2 transformed those findings into a microservices architecture with profile management, skill matching, workspaces, notifications, analytics, and LMS/library integration. Workshop #3 strengthened the design through fault tolerance, risk management, quality assurance, project planning, and the B1 balancing loop for isolation mitigation.

II. CONTINUITY FROM PREVIOUS WORKSHOPS

TABLE I. PROJECT EVOLUTION

W	Output	Simulation use
#1	Survey, AS-IS/TO-BE, loops	Calibration and baseline
#2	Microservices and NFRs	Architecture and latency targets
#3	Risk, QA, B1 loop	Stress/failure scenarios
#4	Models and validation	Empirical design evidence

III. SIMULATION METHODOLOGY

A. Process-Oriented Simulation

The process model represents the sequence from student need recognition to successful group formation. It compares the current WhatsApp-based process with platform-supported matching, notification, schedule integration, LMS fallback, and exam-period stress scenarios. Each scenario is evaluated through Monte Carlo trials using synthetic profiles calibrated from Workshop #1.

B. Behavior-Oriented Simulation

The behavior model represents students as agents in an undirected collaboration graph. Weekly interactions create or reinforce edges, while adoption and satisfaction influence participation. Metrics include density, isolated students, centrality inequality, giant-component coverage, and satisfaction across a 12-week semester.

C. Assumptions

Student skill, availability, performance, social connectivity, and platform openness are normalized synthetic variables. Institutional academic records are not used, preserving privacy while allowing design-level validation.

IV. EXPERIMENTAL DESIGN

TABLE II. SIMULATION SCENARIOS

Scenario	Purpose	Change
AS-IS	Current coordination	No platform
Baseline	Workshop #2 design	80% adoption
Opt. B1	Isolation control	B1 priority
LMS Out.	Risk fallback	LMS unavailable
Peak 500	Capacity test	500 students

V. PROCESS SIMULATION RESULTS

TABLE III. PROCESS RESULTS

Sc.	n	Succ.	Days	p95	Iso.	GBI	Sat.
AS-IS	200	0.39	4.66	N/A	122	0.35	0.43
Base	200	0.91	0.38	0.27	69	0.64	0.87
Opt.	200	0.98	0.40	0.21	40	0.64	0.89
Out.	200	0.74	0.51	0.32	93	0.64	0.77
Peak	500	0.94	0.39	1.24	130	0.62	0.87

The optimized ACN scenario compresses group formation from 4.66 days to 0.40 days and improves satisfaction from 0.43 to 0.89. The LMS outage case remains operational with success rate 0.74, validating the fallback but showing integration reliability still matters.

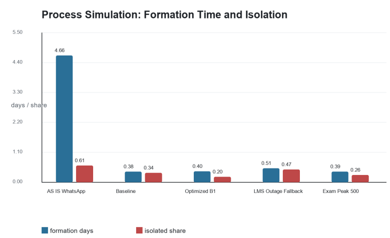

Fig. 1. Formation time and isolated-student share.

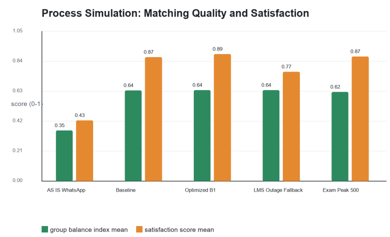

Fig. 2. Matching quality and satisfaction.

VI. BEHAVIOR SIMULATION

TABLE IV. BEHAVIOR RESULTS

Sc.	Ad.	Den.	Iso.	Gini	Giant	Sat.
AS-IS	0.00	0.012	29	0.40	0.82	0.45
No B1	0.51	0.026	7	0.43	0.96	0.63
B1	0.87	0.116	5	0.19	0.97	0.79
Low	0.37	0.026	11	0.47	0.94	0.62

The behavior model shows nonlinear network growth. Informal self-selection ends with 29 isolated students and density 0.012. With B1, final isolation falls to 5, density rises to 0.116, and degree inequality decreases to 0.19.

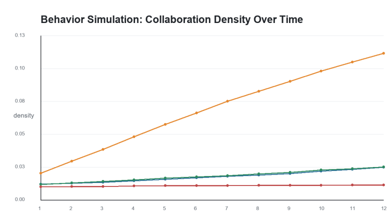

Fig. 3. Collaboration density over 12 weeks.

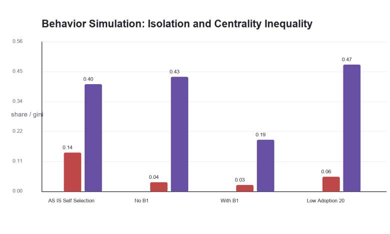

Fig. 4. Isolation and centrality inequality.

VII. COMPLEXITY AND SENSITIVITY

The system exhibits nonlinear adoption effects, centrality concentration in informal groups, and balancing behavior when low-degree students are prioritized. Sensitivity tests confirm that adoption is structural: below a mid-range threshold, the graph does not accumulate enough active participants to eliminate isolation.

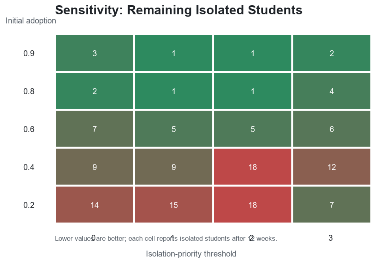

Fig. 5. Sensitivity of remaining isolated students.

VIII. DESIGN VALIDATION

TABLE V. VALIDATION SUMMARY

Decision	Evidence	Result
Matcher	p95 1.24s at 500	Valid
Workspace	Time -91.5%	Valid
B1 loop	Isolation 29->5	Valid
LMS fallback	Success 0.74	Partial
Adoption	Low case leaves 11	Mitigate

IX. RECOMMENDATIONS

- Implement B1 isolation-priority as a mandatory Matching Engine component.
- Use cached pre-filtering and incremental recomputation for sub-three-second matching.
- Launch faculty ambassadors, guided onboarding, and pilot incentives with the MVP.
- Maintain LMS fallback mechanisms and monitor schedule compatibility during outages.
- Apply consent, data minimization, access control, explainability, and bias audits.
- Run a 50-student pilot to replace synthetic assumptions with observed data.

X. LIMITATIONS AND ETHICS

The model uses synthetic profiles calibrated from a limited survey sample. It approximates interpersonal compatibility, motivation, instructor intervention, and LMS behavior. Results should be interpreted as design validation evidence rather than a production forecast. Any implementation must protect student privacy and avoid biased academic recommendations.

XI. CONCLUSION

The simulation phase validates the ACN design trajectory across the four-workshop sequence. The architecture can reduce coordination time, improve satisfaction, preserve latency targets under modeled stress, and reduce isolation when centrality-based re-inclusion is active. The strongest remaining risk is sustained adoption, making change management a technical success factor.

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GITHUB REPOSITORY

Workshop 4 materials should be placed under Workshop_4_Simulation in the course repository, including source code, input data, figures, results, README documentation, and this report. Repository: https://github.com/AnyeloCZ/Academic-Collaboration-Network-SAD-2026-I

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